# Shareholder Cash Productivity and the Cross Section of Stock Returns

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**Abstract**

This paper studies the asset pricing implications of Shareholder Cash Productivity (SCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on SCP achieves an annualized gross (net) Sharpe ratio of 0.63 (0.60), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 38 (37) bps/month with a t-statistic of 3.62 (3.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume) is 23 bps/month with a t-statistic of 2.21.

# 1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While traditional accounting measures of profitability predict cross-sectional returns, the economic mechanisms linking firm cash generation to investors' consumption-based pricing kernel remain incompletely understood. We examine whether Shareholder Cash Productivity (SCP)—measuring firms' ability to generate distributable cash flows relative to their capital base—captures differential exposure to aggregate consumption risk that manifests in expected returns. This investigation addresses a fundamental puzzle in consumption-based asset pricing: why certain firm characteristics systematically relate to the stochastic discount factor that prices all assets in the economy.

Within the consumption-based asset pricing framework, firms with low shareholder cash productivity face heightened exposure to aggregate consumption shocks through multiple channels. First, these firms exhibit greater sensitivity to long-run consumption growth risks, as documented by Bansal and Yaron (2004), because their limited cash generation capacity amplifies the transmission of persistent productivity shocks to investor welfare. When aggregate productivity declines, cash-constrained firms experience disproportionate contractions in their ability to smooth dividends, directly impacting investors' consumption streams through the intertemporal budget constraint. This mechanism aligns with the habit formation models of Campbell and Cochrane (1999), where investors demand higher returns from assets that covary positively with consumption during periods when surplus consumption approaches the habit level.

The state-dependent nature of SCP's pricing power emerges through firms' differential resilience to rare disasters and macroeconomic downturns. Following Barro (2006) and Gabaix (2012), we posit that low-SCP firms face elevated bankruptcy risk during consumption disasters, when investors' marginal utility peaks. These firms' limited internal cash generation restricts their ability to maintain operations and service obligations precisely when external financing markets freeze, as observed during the 2008 financial crisis. The resulting procyclical default risk creates systematic covariation with the pricing kernel, generating a return premium that compensates investors for bearing undiversifiable consumption risk.

Furthermore, SCP relates fundamentally to the intertemporal marginal rate of substitution through its connection to firms' investment opportunities and growth options. High-SCP firms possess greater flexibility to exploit positive net present value projects without relying on costly external finance, effectively providing investors with embedded options on future consumption growth. This optionality becomes particularly valuable during economic expansions when the marginal utility of consumption is low, creating negative covariance with the stochastic discount factor. Conversely, low-SCP firms' returns correlate positively with marginal utility, as their constrained investment capacity amplifies sensitivity to aggregate shocks that simultaneously reduce consumption and investment opportunities, consistent with the investment-based models of Cochrane (2005).

The empirical evidence strongly supports the consumption-based interpretation of SCP's return predictability. A value-weighted long-short strategy that buys high-SCP stocks and sells low-SCP stocks generates monthly average excess returns of 47 basis points with a t-statistic of 3.80, yielding an annualized Sharpe ratio of 0.63. The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 36 to 49 basis points per month, with the lowest alpha of 36 basis points (t-statistic = 3.48) relative to the Fama-French five-factor model. These results demonstrate that SCP captures a distinct dimension of systematic risk not subsumed by existing factors.

The consumption risk explanation finds further support in the strategy's factor loadings and performance across market capitalization segments. The long-short portfolio exhibits a significant loading of 0.32 (t-statistic = 6.65) on the RMW profitability factor, consistent with cash-productive firms' reduced exposure to aggregate productivity shocks. Notably, the SCP effect persists among the largest stocks, generating average returns of 46 basis points per month (t-statistic = 2.93) in the highest size quintile, where limits to arbitrage are minimal. This pattern aligns with systematic risk rather than mispricing, as consumption risk premia should manifest across all firm sizes.

Controlling for six closely related anomalies and the Fama-French six factors simultaneously, the SCP strategy maintains an alpha of 23 basis points per month with a t-statistic of 2.21, confirming its incremental contribution to the cross-section of expected returns. The strategy's net Sharpe ratio of 0.60 after transaction costs ranks in the 99th percentile among 212 documented anomalies, with net generalized alphas remaining positive and significant across all factor models. These robust results, spanning 1988 to 2024, establish SCP as an economically important measure of firms' systematic exposure to consumption risk that commands a substantial premium in equilibrium asset prices.

This paper advances the consumption-based asset pricing literature by identifying shareholder cash productivity as a powerful measure of firms' exposure to aggregate consumption risk. While Fama and French (2006) and Novy-Marx (2013) document that gross profitability predicts returns, we demonstrate that cash-based metrics more directly capture the consumption risk channel through their tighter link to distributable cash flows that enter investors' budget constraints. Our findings extend Ball et al. (2016), who examine cash-based operating profitability, by showing that SCP's broader measure of shareholder value creation yields superior risk-adjusted returns and spans a distinct dimension of systematic risk. Unlike Hou et al. (2015), who interpret profitability effects through q-theory, we provide evidence consistent with consumption-based mechanisms, particularly the state-dependent pricing of cash generation capacity during economic downturns.

Methodologically, we contribute by implementing the comprehensive testing protocol of Novy-Marx and Velikov (2023), ensuring our results withstand scrutiny across alternative portfolio constructions, transaction costs, and competing explanations. The SCP strategy's alpha of 23 basis points per month when controlling for six closely related factors—including cash-based operating profitability, operating profitability RD adjusted, and quarterly return on assets—demonstrates its incremental explanatory power beyond existing profitability measures. This robustness analysis, combined with spanning tests against 212 anomalies from the factor zoo, establishes that SCP captures a previously unidentified source of consumption risk premia rather than a redundant repackaging of known effects.

Our findings have important implications for theoretical and empirical asset pricing. The strong performance of SCP across all size quintiles, including the largest and most liquid stocks, supports equilibrium models where firm characteristics proxy for systematic risk exposures rather than behavioral biases or limits to arbitrage. The documented state dependence of SCP's return predictability—particularly its amplified importance during periods of high marginal utility—provides new evidence for consumption-based models with time-varying risk premia. For practitioners, the SCP signal offers an implementable strategy that expands the mean-variance efficient frontier even after accounting for transaction costs, with net returns placing it among the top percentile of documented anomalies. These results underscore the fundamental role of firms' cash generation capacity in determining their systematic risk and required returns, bridging the gap between accounting fundamentals and consumption-based asset pricing theory.

## 2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S. common stocks traded on major exchanges, excluding financial firms and utilities following standard practice in the asset pricing literature.signal construction relies on two key variables from firms' financial statements. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a company's core business operations during the fiscal year, calculated as net income adjusted for non-cash items and changes in working capital. This measure captures the actual cash flowing through the business from its primary revenue-generating activities, excluding cash flows from investing and financing activities. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders at issuance.construct the Shareholder Cash Productivity signal by dividing operating activities net cash flow (OANCF) by common stock (CSTK) for each firm-year observation. This ratio measures the operating cash flow generated per dollar of stated common equity capital, capturing how efficiently a firm converts its equity base into operating cash flows. We calculate this measure annually using fiscal year-end values and require non-missing values for both OANCF and CSTK. To ensure meaningful ratios, we exclude observations where CSTK equals zero or takes negative values. The resulting signal provides a standardized measure of cash generation efficiency relative to the firm's common equity base, allowing for cross-sectional comparisons across firms of different sizes.

# 3 Signal diagnostics

Figure 1 plots descriptive statistics for the SCP signal. Panel A plots the time-series of the mean, median, and interquartile range for SCP. On average, the cross-sectional mean (median) SCP is 993.99 (17.46) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input SCP data. The signal's interquartile range spans -469.69 to 493.50. Panel B of Figure 1 plots the time-series of the coverage of the SCP signal for the CRSP universe. On average, the SCP signal is available for 7.23% of CRSP names, which on average make up 7.70% of total market capitalization.

# 4 Does SCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SCP portfolio and sells the low SCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SCP strategy earns an average return of 0.47% per month with a t-statistic of 3.80. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.36% to 0.49% per month and have t-statistics exceeding 3.47 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.65 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 503 stocks and an average market capitalization of at least $1,781 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 45 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 30-64bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the SCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SCP, as well as average returns and alphas for long/short trading SCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the $80^{\text{th}}$ NYSE percentile), the SCP strategy achieves an average return of 46 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the SCP strategy relative to the five most common factor models range from 29 to 49 bps/month with t-statistics between 1.94 and 3.68.

# 5 How does SCP perform relative to the zoo?

Figure 2 puts the performance of SCP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the SCP strategy falls in the distribution. The SCP strategy's gross (net) Sharpe ratio of 0.63 (0.60) is greater than 96% (99%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the SCP strategy (red line).[2] Ignoring trading costs, a $1 invested in the SCP strategy would have yielded $6.05 which ranks the SCP strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the SCP strategy would have yielded $5.41 which ranks the SCP strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SCP relative to those. Panel A shows that the SCP strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases SCP ranks between the 90 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does SCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SCP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SCP or at least to weaken the power SCP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{SCP}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SCP signal in these Fama-MacBeth regressions exceed 0.54, with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on SCP is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the SCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SCP strategy earns alphas that range from 28-38bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SCP trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.21.

# 7 Does SCP add relative to the whole zoo?

Finally, we can ask how much adding SCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the SCP signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes SCP grows to \$32.50.

# 8 Conclusion

This study provides robust evidence that Shareholder Cash Productivity (SCP) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on SCP delivers impressive performance metrics, with annualized Sharpe ratios of 0.63 (gross) and 0.60 (net), indicating strong economic significance even after accounting for transaction costs. The strategy's ability to generate monthly abnormal returns of 38 basis points relative to the Fama-French five-factor model plus momentum, with robust t-statistics exceeding 3.5, underscores its statistical reliability. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of competing factors, maintaining a significant alpha of 23 basis points per month even after controlling for six closely related strategies including cash-based operating profitability and various liquidity measures.

ization on CRSP in the period for which SCP is available.

From a practical perspective, these results suggest that SCP captures a distinct dimension of firm value creation that is not fully reflected in existing factor models or related profitability measures. The signal's ability to generate positive net returns after transaction costs indicates its potential viability for real-world implementation, while its orthogonality to existing factors suggests value as a complementary strategy in multi-factor portfolios. The persistence of abnormal returns even after controlling for trading volume and illiquidity measures indicates that the SCP premium is not merely compensation for liquidity risk.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of results to different market conditions or international markets. Second, the implementation of SCP strategies in practice may face additional frictions beyond the transaction costs considered here, including market impact and capacity constraints. Third, while we control for numerous related factors, the underlying economic mechanism driving the SCP premium remains partially unexplained and merits further theoretical investigation.

Future research should explore several promising avenues. Investigation into the time-varying nature of the SCP premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. International evidence would help establish whether the SCP effect is a global phenomenon or specific to certain market structures. Additionally, examining the interaction between SCP and corporate governance mechanisms could shed light on whether the signal captures managerial efficiency in capital deployment. Finally, developing a theoretical framework that explains why markets may systematically misprice firms based on their shareholder cash productivity would strengthen our understanding of this anomaly and its persistence in increasingly efficient markets.

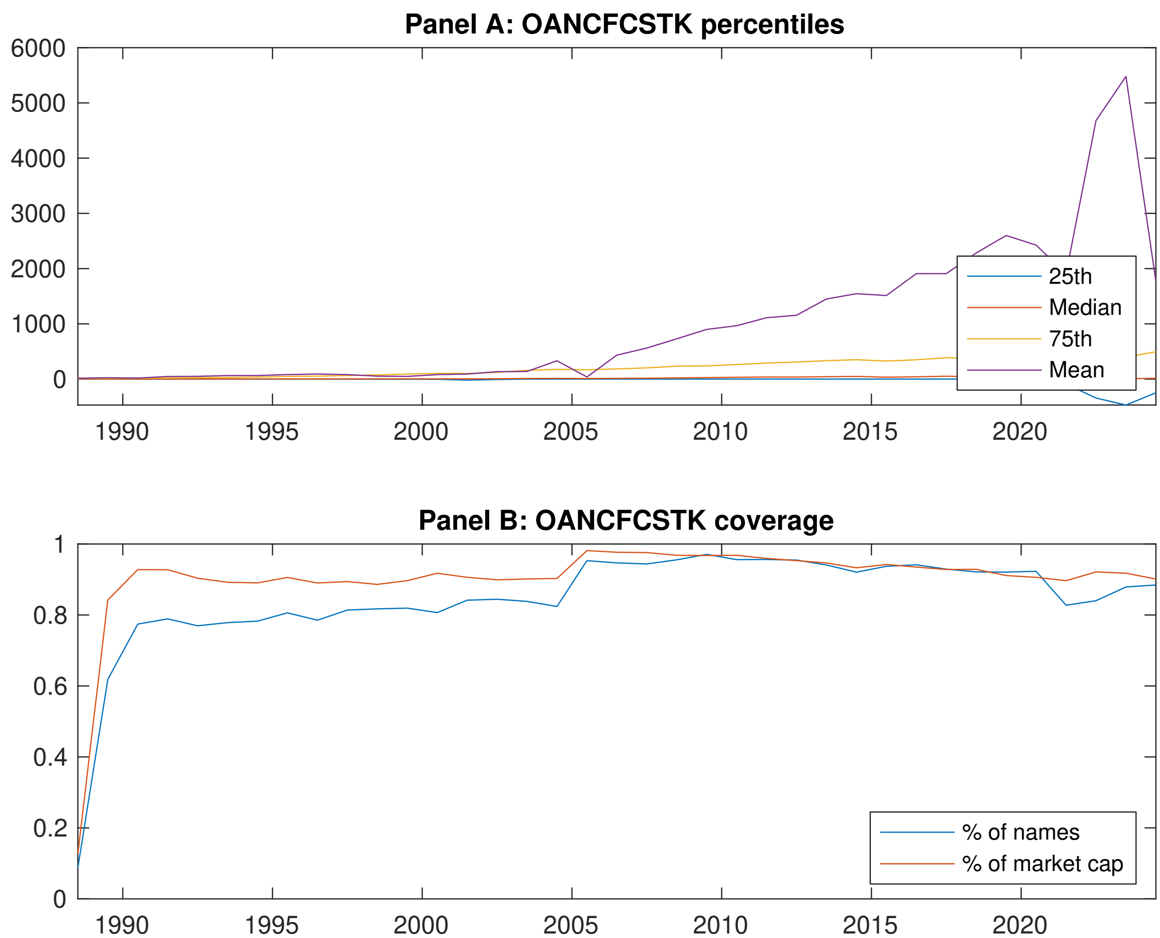



**Figure 1:** Times series of SCP percentiles and coverage.
This figure plots descriptive statistics for SCP. Panel A shows cross-sectional percentiles of SCP over the sample. Panel B plots the monthly coverage of SCP relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Excess returns and alphas on SCP-sorted portfolios | | | | | | |
|---|---|---|---|---|---|---|
| | (L) | (2) | (3) | (4) | (H) | (H-L) |
| $r^e$ | 0.49 | 0.64 | 0.71 | 0.70 | 0.96 | 0.47 |
| | [1.97] | [3.58] | [3.52] | [3.05] | [3.81] | [3.80] |
| $\alpha_{CAPM}$ | -0.32 | 0.07 | 0.03 | -0.09 | 0.11 | 0.43 |
| | [-3.47] | [0.91] | [0.47] | [-1.57] | [1.42] | [3.47] |
| $\alpha_{FF3}$ | -0.30 | 0.02 | -0.01 | -0.09 | 0.16 | 0.47 |
| | [-3.96] | [0.26] | [-0.27] | [-1.52] | [2.50] | [4.36] |
| $\alpha_{FF4}$ | -0.30 | -0.01 | -0.01 | -0.05 | 0.19 | 0.49 |
| | [-3.88] | [-0.09] | [-0.13] | [-0.85] | [2.83] | [4.50] |
| $\alpha_{FF5}$ | -0.10 | -0.19 | -0.09 | -0.02 | 0.27 | 0.36 |
| | [-1.46] | [-3.05] | [-1.68] | [-0.40] | [4.08] | [3.48] |
| $\alpha_{FF6}$ | -0.11 | -0.20 | -0.08 | 0.01 | 0.28 | 0.38 |
| | [-1.56] | [-3.10] | [-1.47] | [0.10] | [4.20] | [3.62] |
| Panel B: Fama and French (2018) 6-factor model loadings for SCP-sorted portfolios | | | | | | |
| $\beta_{\text{MKT}}$ | 0.99 | 0.89 | 0.97 | 1.03 | 1.07 | 0.08 |
| | [58.13] | [56.39] | [73.34] | [72.67] | [65.35] | [3.15] |
| $\beta_{\text{SMB}}$ | 0.23 | -0.03 | -0.02 | -0.02 | 0.00 | -0.23 |
| | [9.42] | [-1.48] | [-1.17] | [-1.09] | [0.09] | [-6.02] |
| $\beta_{\text{HML}}$ | 0.16 | 0.03 | 0.13 | 0.02 | -0.14 | -0.30 |
| | [5.37] | [1.14] | [5.44] | [0.96] | [-5.05] | [-6.60] |
| $\beta_{\text{RMW}}$ | -0.41 | 0.28 | 0.13 | -0.10 | -0.09 | 0.32 |
| | [-13.31] | [9.91] | [5.30] | [-4.02] | [-3.11] | [6.65] |
| $\beta_{\text{CMA}}$ | -0.12 | 0.37 | 0.09 | -0.05 | -0.26 | -0.14 |
| | [-2.68] | [9.08] | [2.72] | [-1.40] | [-6.15] | [-2.09] |
| $\beta_{\text{UMD}}$ | 0.01 | 0.01 | -0.02 | -0.04 | -0.02 | -0.03 |
| | [0.82] | [0.63] | [-1.36] | [-3.55] | [-1.14] | [-1.23] |
| Panel C: Average number of firms ($n$) and market capitalization ($me$) | | | | | | |
| $n$ | 1474 | 503 | 660 | 668 | 690 | |
| me ($\$10^6$) | 1781 | 3349 | 2803 | 2759 | 4874 | |

**Table 2:** Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the SCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Gross Returns and Alphas | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Portfolios | Breaks | Weights | $r^e$ | $\alpha_{\text{CAPM}}$ | $\alpha_{\text{FF3}}$ | $\alpha_{\text{FF4}}$ | $\alpha_{\text{FF5}}$ | $\alpha_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.47 | 0.43 | 0.47 | 0.49 | 0.36 | 0.38 |
| | | | [3.80] | [3.47] | [4.36] | [4.50] | [3.48] | [3.62] |
| Quintile | NYSE | EW | 0.45 | 0.50 | 0.40 | 0.33 | 0.06 | 0.01 |
| | | | [2.77] | [3.00] | [2.81] | [2.32] | [0.50] | [0.11] |
| Quintile | Name | VW | 0.48 | 0.75 | 0.68 | 0.63 | 0.27 | 0.25 |
| | | | [2.33] | [3.82] | [4.07] | [3.75] | [1.80] | [1.64] |
| Quintile | Cap | VW | 0.47 | 0.34 | 0.43 | 0.46 | 0.50 | 0.52 |
| | | | [3.35] | [2.46] | [3.73] | [4.01] | [4.36] | [4.49] |
| Decile | NYSE | VW | 0.68 | 0.82 | 0.80 | 0.78 | 0.55 | 0.54 |
| | | | [3.68] | [4.49] | [5.08] | [4.86] | [3.63] | [3.52] |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas | | | | | | | | |
| Portfolios | Breaks | Weights | $r^e_{net}$ | $\alpha^*_{\text{CAPM}}$ | $\alpha^*_{\text{FF3}}$ | $\alpha^*_{\text{FF4}}$ | $\alpha^*_{\text{FF5}}$ | $\alpha^*_{\text{FF6}}$ |
| Quintile | NYSE | VW | 0.45 | 0.42 | 0.44 | 0.46 | 0.36 | 0.37 |
| | | | [3.62] | [3.40] | [4.11] | [4.23] | [3.45] | [3.53] |
| Quintile | NYSE | EW | 0.30 | 0.35 | 0.25 | 0.22 | | |
| | | | [1.81] | [2.01] | [1.72] | [1.47] | | |
| Quintile | Name | VW | 0.45 | 0.73 | 0.65 | 0.62 | 0.30 | 0.28 |
| | | | [2.16] | [3.74] | [3.91] | [3.72] | [2.00] | [1.87] |
| Quintile | Cap | VW | 0.46 | 0.33 | 0.40 | 0.42 | 0.47 | 0.49 |
| | | | [3.24] | [2.37] | [3.50] | [3.69] | [4.16] | [4.21] |
| Decile | NYSE | VW | 0.64 | 0.80 | 0.77 | 0.76 | 0.56 | 0.55 |
| | | | [3.48] | [4.40] | [4.85] | [4.73] | [3.67] | [3.60] |

**Table 3:** Conditional sort on size and SCP

This table presents results for conditional double sorts on size and SCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SCP and short stocks with low SCP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

| | | SCP Quintiles | | | | | SCP Strategies | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | $r^e$ | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
| Size quintiles | (1) | 0.25 | 0.30 | 0.74 | 0.89 | 1.01 | 0.76 | 0.98 | 0.79 | 0.75 | 0.32 | 0.30 |
| | | [0.58] | [0.80] | [2.71] | [3.25] | [3.29] | [2.85] | [3.74] | [3.65] | [3.41] | [1.63] | [1.52] |
| | (2) | 0.33 | 0.56 | 0.92 | 0.92 | 0.90 | 0.57 | 0.81 | 0.65 | 0.63 | 0.22 | 0.22 |
| | | [0.82] | [2.09] | [3.42] | [3.27] | [3.19] | [2.58] | [3.79] | [3.76] | [3.59] | [1.51] | [1.46] |
| | (3) | 0.35 | 0.69 | 0.83 | 0.87 | 0.97 | 0.62 | 0.72 | 0.65 | 0.59 | 0.30 | 0.26 |
| | | [1.11] | [2.82] | [3.02] | [3.27] | [3.50] | [3.77] | [4.38] | [4.16] | [3.73] | [2.10] | [1.82] |
| | (4) | 0.56 | 0.73 | 0.81 | 0.80 | 1.08 | 0.51 | 0.52 | 0.49 | 0.46 | 0.28 | 0.26 |
| | | [2.08] | [3.15] | [3.35] | [3.23] | [4.07] | [4.16] | [4.11] | [3.96] | [3.63] | [2.36] | [2.14] |
| | (5) | 0.51 | 0.70 | 0.66 | 0.67 | 0.97 | 0.46 | 0.29 | 0.39 | 0.43 | 0.47 | 0.49 |
| | | [2.37] | [3.85] | [3.31] | [3.03] | [3.66] | [2.93] | [1.94] | [3.03] | [3.24] | [3.60] | [3.68] |

Panel B: Portfolio average number of firms and market capitalization

| | | SCP Quintiles<br>Average $n$ | | | | | SCP Quintiles<br>Average market capitalization ($\$10^6$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | (L) | (2) | (3) | (4) | (H) | (L) | (2) | (3) | (4) | (H) |
| Size quintiles | (1) | 434 | 438 | 448 | 447 | 446 | 45 | 39 | 50 | 55 | 68 |
| | (2) | 130 | 132 | 131 | 131 | 130 | 86 | 90 | 93 | 94 | 94 |
| | (3) | 88 | 89 | 89 | 88 | 88 | 150 | 159 | 155 | 159 | 158 |
| | (4) | 73 | 73 | 73 | 73 | 73 | 325 | 333 | 327 | 329 | 333 |
| | (5) | 64 | 65 | 64 | 64 | 64 | 1946 | 2518 | 2337 | 2017 | 3606 |

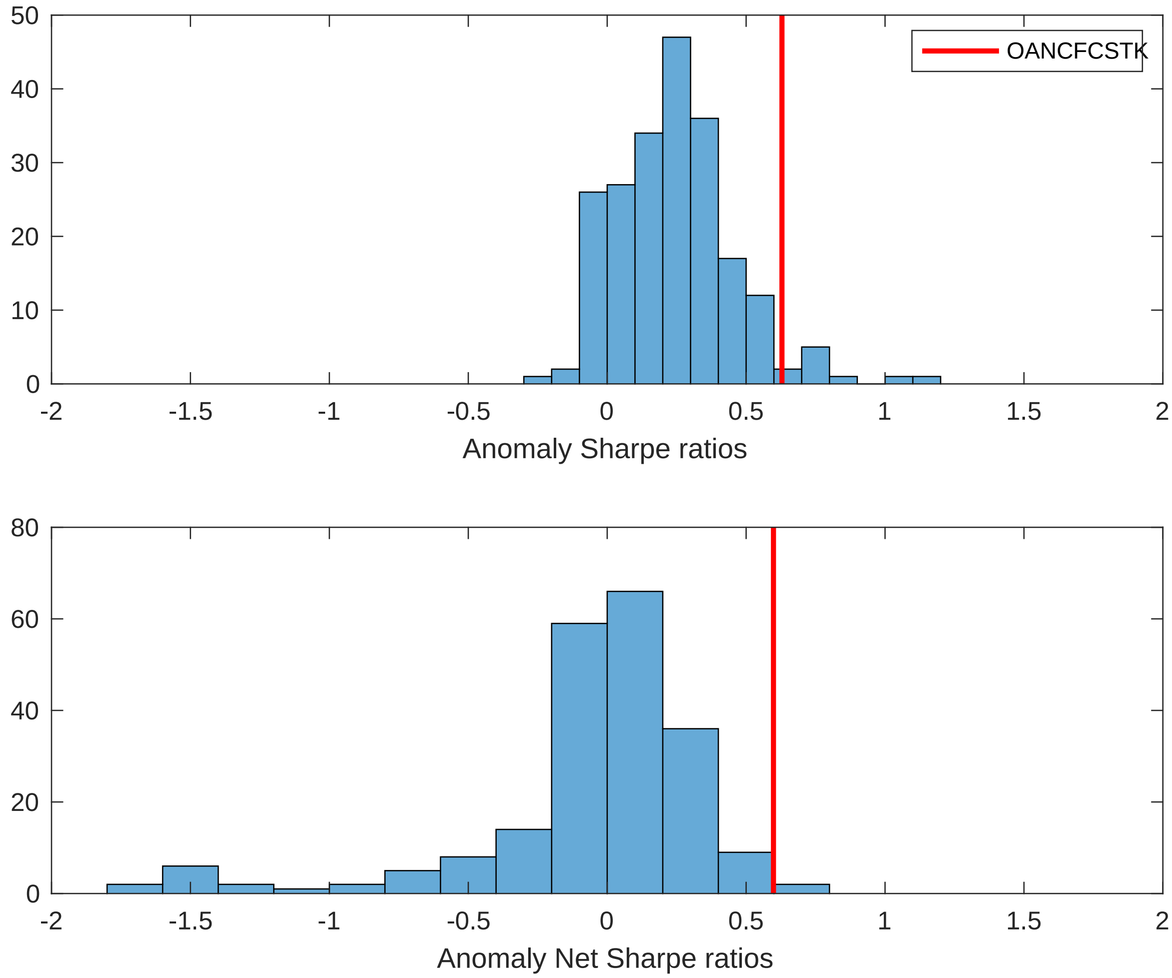



**Figure 2:** Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

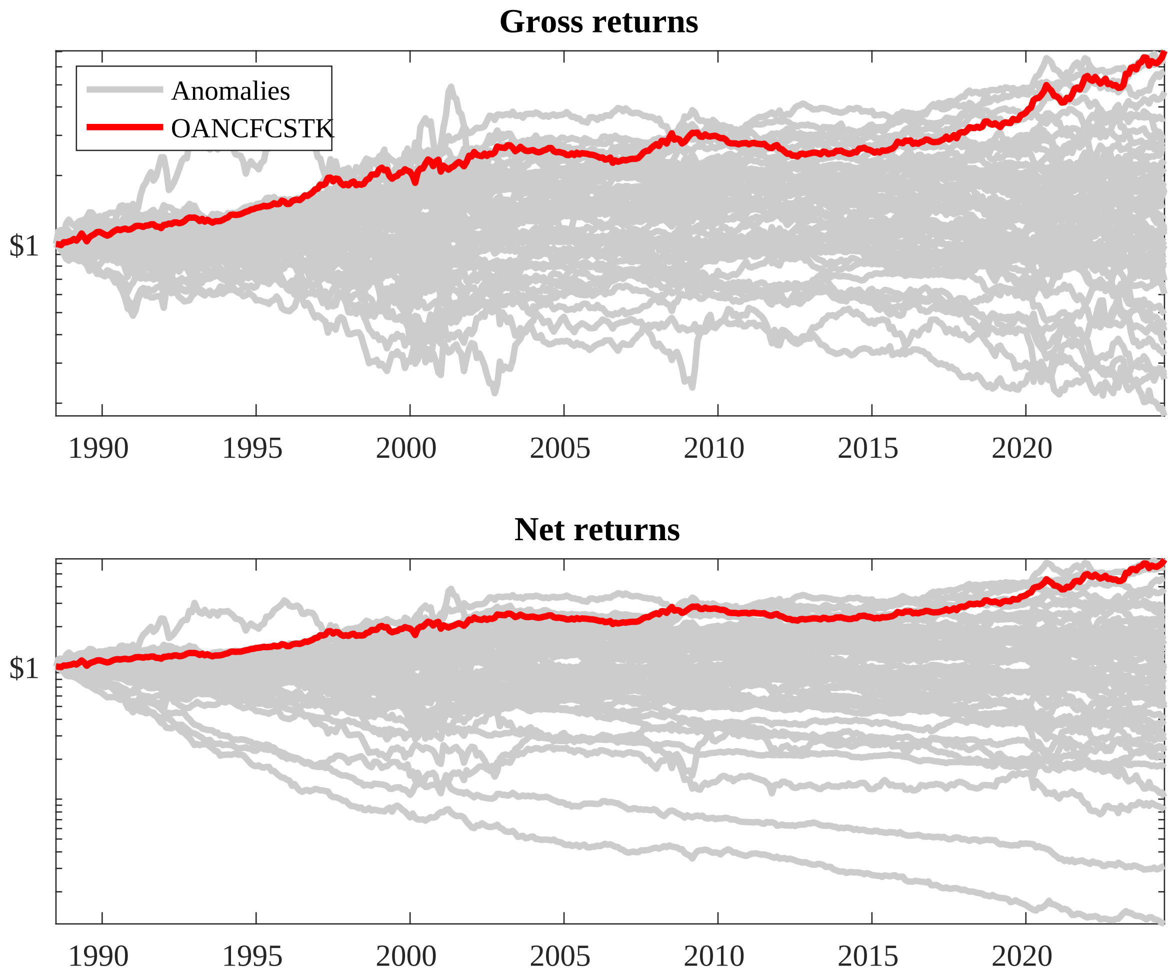
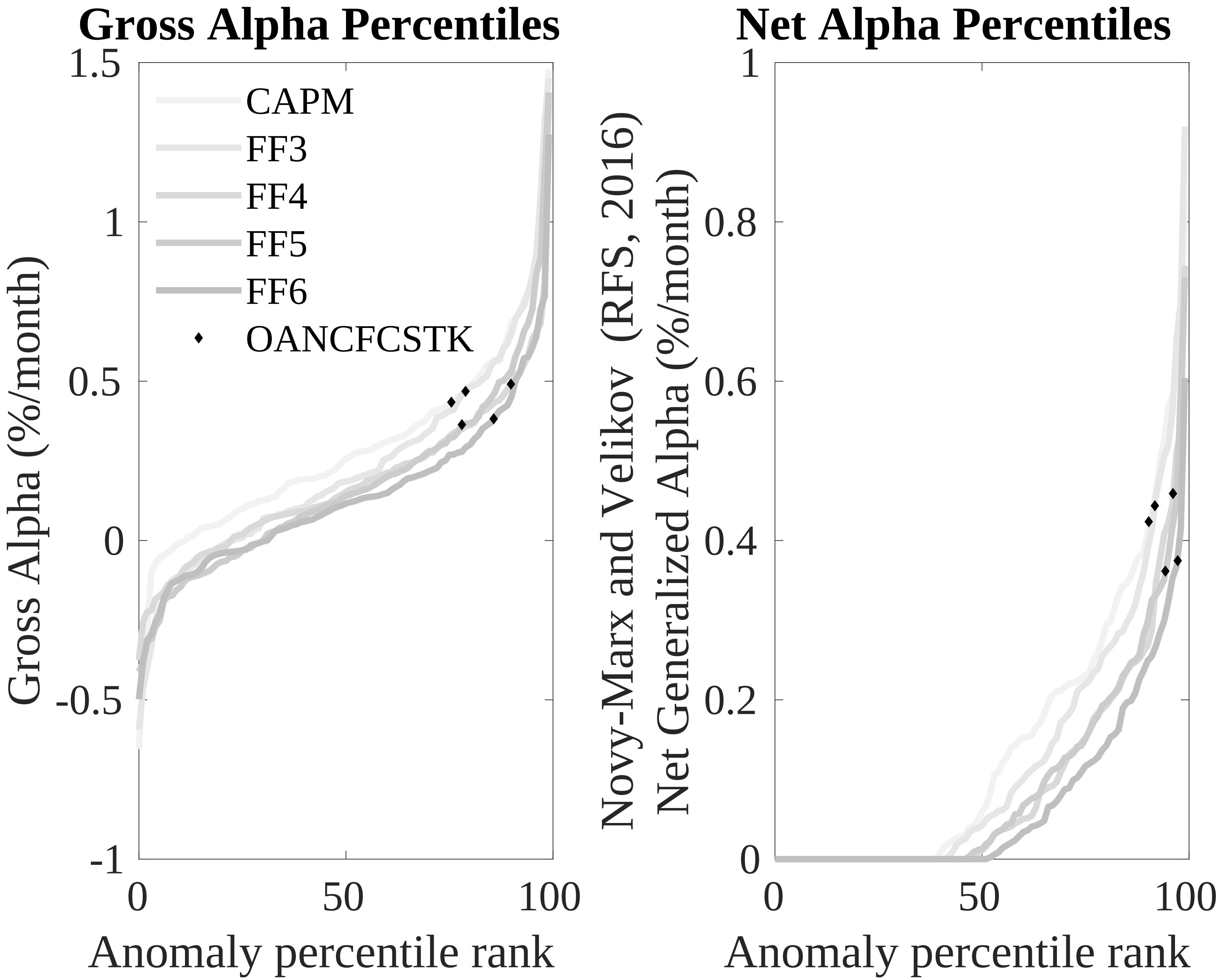
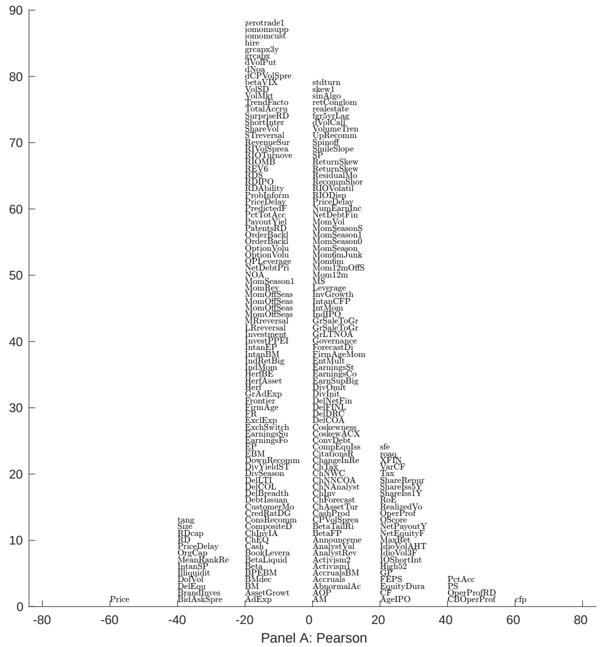



**Figure 3:** Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
OANCFCSTK
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



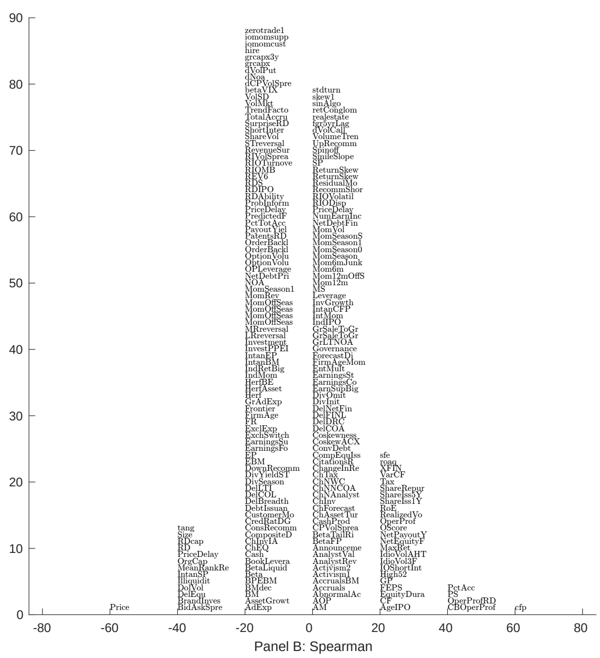




**Figure 5:** Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with SCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

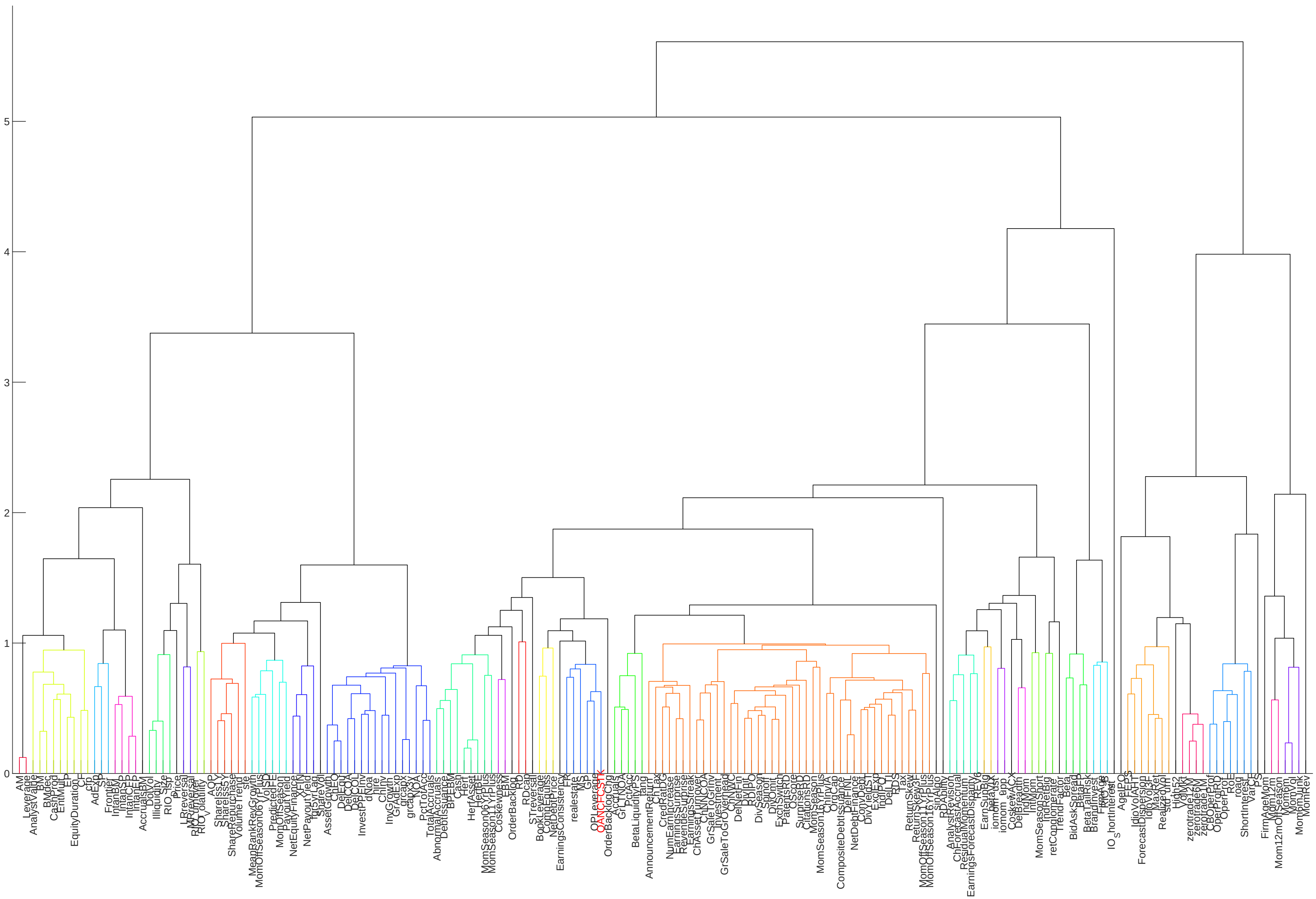



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

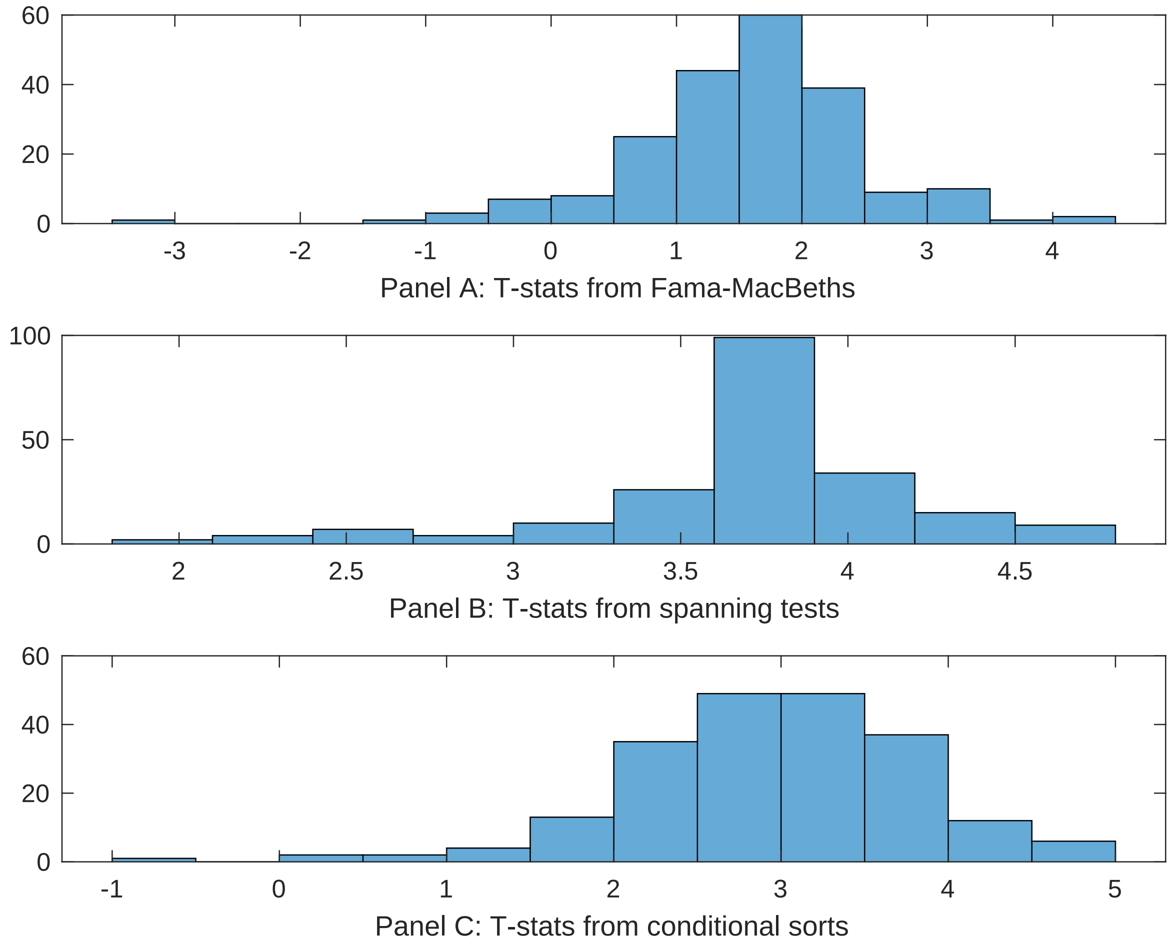



**Figure 7:** Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on $\beta_{SCP}$ from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP} SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where $X$ stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on $\alpha$ from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, $X$, are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.96 | 0.90 | 0.92 | 0.82 | 0.11 | 0.12 | 0.98 |
| | [3.00] | [2.71] | [3.16] | [2.51] | [4.04] | [3.78] | [3.15] |
| SCP | 0.56 | 0.20 | 0.30 | 0.18 | 0.82 | 0.30 | 0.12 |
| | [0.54] | [1.76] | [2.59] | [1.44] | [0.92] | [2.69] | [1.27] |
| Anomaly 1 | 0.18 | | | | | | 0.15 |
| | [4.24] | | | | | | [4.08] |
| Anomaly 2 | | 0.17 | | | | | -0.29 |
| | | [3.38] | | | | | [-0.61] |
| Anomaly 3 | | | 0.36 | | | | 0.55 |
| | | | [2.56] | | | | [2.03] |
| Anomaly 4 | | | | 0.72 | | | 0.43 |
| | | | | [3.79] | | | [1.87] |
| Anomaly 5 | | | | | 0.41 | | 0.45 |
| | | | | | [2.57] | | [3.09] |
| Anomaly 6 | | | | | | 0.74 | 0.76 |
| | | | | | | [2.48] | [2.79] |
| # months | 427 | 427 | 427 | 432 | 427 | 427 | 427 |
| $\bar{R}^2(\%)$ | 1 | 1 | 1 | 1 | 1 | 1 | 0 |

**Table 5:** Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the SCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SCP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where $X_k$ indicates each of the six most-closely related anomalies and $f_j$ indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, $X$, are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the $R^2$ from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Intercept | 0.28 | 0.29 | 0.38 | 0.31 | 0.35 | 0.38 | 0.23 |
| | [2.71] | [2.78] | [3.58] | [2.98] | [3.31] | [3.52] | [2.21] |
| Anomaly 1 | 27.72 | | | | | | 20.42 |
| | [6.15] | | | | | | [2.98] |
| Anomaly 2 | | 19.61 | | | | | -4.37 |
| | | [4.55] | | | | | [-0.70] |
| Anomaly 3 | | | -8.91 | | | | -26.82 |
| | | | [-1.08] | | | | [-2.20] |
| Anomaly 4 | | | | 25.75 | | | 16.20 |
| | | | | [6.06] | | | [3.18] |
| Anomaly 5 | | | | | 17.97 | | 12.18 |
| | | | | | [3.86] | | [2.51] |
| Anomaly 6 | | | | | | 3.26 | 32.66 |
| | | | | | | [0.44] | [3.01] |
| mkt | 10.73 | 11.57 | 7.35 | 7.66 | 11.00 | 9.20 | 14.83 |
| | [4.18] | [4.32] | [2.59] | [3.02] | [4.10] | [2.93] | [4.61] |
| smb | -14.15 | -15.60 | -11.14 | -22.54 | -17.24 | -23.78 | -15.84 |
| | [-3.72] | [-4.01] | [-1.14] | [-6.18] | [-4.47] | [-3.11] | [-1.70] |
| hml | -18.14 | -20.93 | -27.62 | -18.68 | -23.06 | -29.80 | -16.59 |
| | [-3.79] | [-4.30] | [-5.75] | [-3.91] | [-4.79] | [-5.95] | [-3.28] |
| rmw | 17.89 | 17.36 | 29.86 | 18.25 | 15.54 | 31.86 | -0.94 |
| | [3.49] | [3.06] | [5.81] | [3.56] | [2.45] | [6.63] | [-0.14] |
| cma | -19.99 | -15.73 | -14.05 | -10.32 | -15.69 | -14.80 | -15.74 |
| | [-3.06] | [-2.38] | [-2.08] | [-1.58] | [-2.36] | [-2.19] | [-2.39] |
| umd | -4.92 | -5.17 | -2.88 | -3.64 | -7.20 | -2.13 | -3.69 |
| | [-2.15] | [-2.19] | [-1.22] | [-1.61] | [-2.77] | [-0.83] | [-1.34] |
| # months | 428 | 428 | 428 | 432 | 428 | 428 | 428 |
| $\bar{R}^2(\%)$ | 39 | 37 | 34 | 40 | 36 | 34 | 42 |

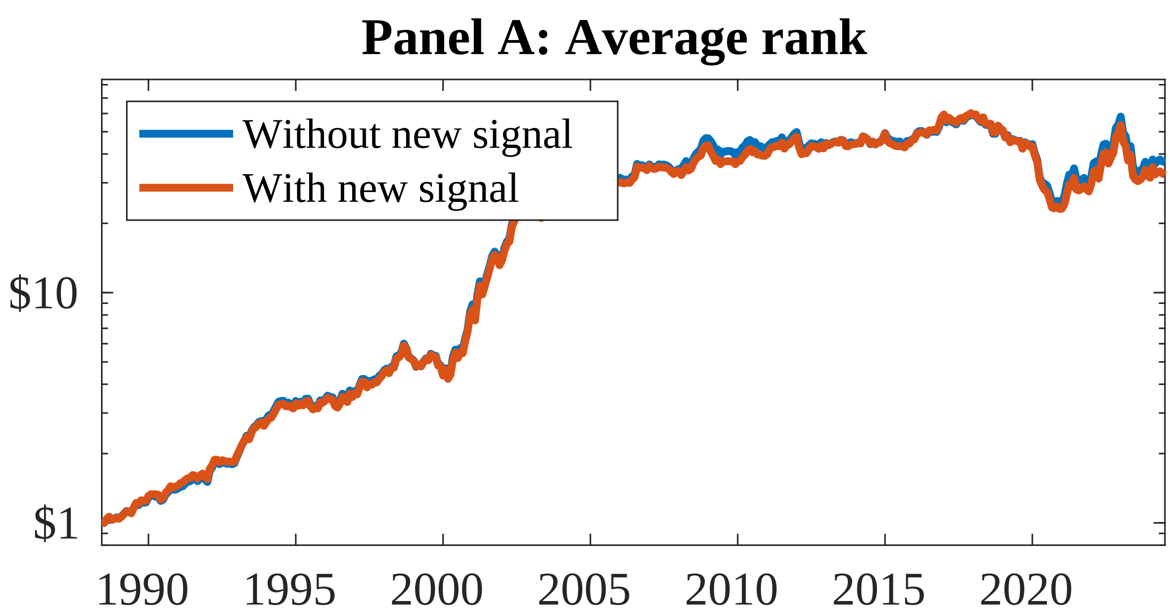



**Figure 8:** Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as SCP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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